

**DESIGN AND IMPLEMENTATION OF A BLOCKCHAIN FOR E-VOTING
USING NATIONAL IDENTIFICATION NUMBER (NIN)**

BY

UGBAH-NDUKA DAVID KING

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DECLARATION

I, **UGBAH-NDUKA DAVID KING** with the Matriculation Number **BHU/22/04/05/0078** declare that this Project was carried out by me and every other external work used in this report has been fully acknowledged.

.....

BHU/22/04/05/0078

UGBAH-NDUKA DAVID KING

.....

DATE

CERTIFICATION

This is to certify that this project was written by **UGBAH-NDUKA DAVID KING** with Matric Number **BHU/22/04/05/0078** and submitted to the Department of Computer Science, Faculty of Computing, Bingham University, Karu, Nigeria, as part of the requirements for the award of **Bachelor of Science (B. Sc.) in Computer Science**.

Mr. Nathan A. Zoakah
Supervisor

DATE

Dr. Adamu S. Usman
Head of Department

DATE

External Examiner

DATE

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DEDICATION

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CHAPTER ONE

INTRODUCTION

1.1 Background of Study

Elections form the foundation of democratic governance, enabling citizens to choose leaders and influence public policy. However, traditional voting methods in Nigeria face serious challenges that undermine electoral credibility and public trust. During the 2023 general elections, the Bimodal Voter Accreditation System (BVAS) experienced failures in approximately 14% of polling units, causing delays, voter disenfranchisement, and voting suspensions (Ene Obi, 2023; olufemiajasa, 2023). Additional problems included late poll openings, missing materials, vote-buying, ballot-box snatching, violence, and failure to upload results (Ene Obi, 2023).

Electronic voting (e-voting) systems offer digital platforms for casting, counting, and transmitting votes efficiently. E-voting aims to improve efficiency, enhance transparency, reduce human error, and strengthen public confidence in electoral outcomes (Adesiyani, 2023). Estonia successfully implemented internet-based voting in 2005 using national ID cards for secure authentication, demonstrating that digital platforms can increase efficiency whilst maintaining security (Fabian Breuer and Alexander H. Trechse, 2006; David Mac Dougall, 2023).

However, conventional e-voting systems face significant limitations. Centralized architectures are vulnerable to cyberattacks, data manipulation, and limited auditability. Poor infrastructure, inadequate training, and connectivity issues further compromise effectiveness in Nigeria (Ene Obi, 2023).

Blockchain technology addresses these limitations through decentralized distributed ledgers that ensure immutability, transparency, and tamper-resistance. Blockchain-based e-voting systems enable verifiable vote records, prevent double voting, preserve voter anonymity, and provide auditable election trails (Russo et al., 2021; Ohize et al., 2024).

Integrating blockchain e-voting with Nigeria's National Identification Number (NIN) strengthens voter authentication and reduces impersonation. The NIN, issued by the National Identity Management Commission (NIMC), provides unique identifiers for Nigerian citizens, enabling secure voter verification (Adesiyani, 2023).

This study designs and implements a blockchain-enabled e-voting platform linked to NIN to address Nigeria's recurring electoral challenges including technical failures, authentication issues, and public mistrust in the Independent National Electoral Commission (INEC). The system aims to enhance the integrity and reliability of Nigerian electoral processes (Ene Obi, 2023; Adesiyan, 2023).

1.2 Statement of Problem

Elections in Nigeria are central to democratic governance, yet persistent challenges undermine their credibility. During the 2023 general elections, approximately 14% of polling units experienced failures in the Bimodal Voter Accreditation System (BVAS), leading to delayed voting, disenfranchisement, and suspension of voting at affected units (Ene Obi, 2023). Election observers reported missing materials, late opening of polling units, ballot-box snatching, intimidation, and failure to upload results for multiple locations (Ene Obi, 2023; olufemiajasa, 2023).

While electronic and biometric systems like BVAS have been introduced to improve efficiency and transparency, centralized architectures remain vulnerable to technical failures, manipulation, and limited auditability. Connectivity issues, power outages, and human error continue to compromise system reliability (Adesiyan, 2023).

Despite these interventions, no existing blockchain-based system integrated with the National Identification Number (NIN) exists in Nigeria to ensure secure, verifiable, and tamper-proof elections. This gap highlights the need for an innovative solution combining decentralized ledger technology with NIN authentication to guarantee transparency, auditability, and trust in electoral outcomes. This study designs and implements a blockchain-enabled e-voting system using NIN to address these systemic deficiencies and enhance the integrity of future Nigerian elections (Russo et al., 2021; Ene Obi, 2023).

1.3 Research Questions

Based on the recurring failures of Nigeria's electoral system and the limitations of current electronic and biometric voting technologies, this study addresses the following research questions:

1. How can a blockchain-based system be designed to ensure secure, verifiable, and tamper-proof e-voting in Nigeria?
2. In what ways can integration with the National Identification Number (NIN) enhance voter authentication and prevent impersonation?
3. What are the technical and operational challenges of implementing blockchain-enabled e-voting in the Nigerian electoral context?
4. How can the proposed system improve transparency, auditability, and public trust compared to existing electronic voting solutions?

1.4 Aim and Objectives of the Study

The aim of this study is to design and implement a secure, transparent, and verifiable blockchain based e-voting system that integrates Nigeria's National Identification Number (NIN) for efficient voter authentication and election integrity. The Objectives of the study are to;

- i. Analyze the limitations of existing electronic and biometric voting systems in Nigeria.
- ii. Design a blockchain enabled e-voting architecture that ensures immutability, auditability, and secure vote recording.
- iii. Integrate/ simulate the system with NIN for robust voter authentication and prevention of impersonation.
- iv. Implement the designed system and conduct functional testing to ensure reliability and security.
- v. Evaluate the system's performance and compare it with existing electronic voting solutions in terms of transparency, efficiency, and user trust.

1.5 Significance of the Study

The significance of this study lies in its potential to enhance the credibility, transparency, and efficiency of elections in Nigeria. By designing a blockchain-based e-voting system integrated with the National Identification Number (NIN), the study aims to:

- i. Reduce electoral fraud and prevent voter impersonation through secure digital voter authentication.
- ii. Improve public trust in the electoral process by ensuring verifiable and tamper-proof vote records.

- iii. Provide a framework for transparent and auditable elections that can minimize disputes and delays in result collation.
- iv. Offer insights and a practical model for policymakers, electoral bodies, and researchers interested in secure and technology-driven voting systems.

1.6 Scope of the Study

The scope of this study is limited to the design and implementation of a blockchain-based e-voting system integrated with Nigeria's National Identification Number (NIN) for secure voter authentication. The study focuses on:

- i. **Development of a Digital Voting Platform:** Designing a user-friendly interface that allows registered voters to cast ballots securely from digital devices.
- ii. **Blockchain Implementation:** Integrating a decentralized ledger to ensure votes are immutable (cannot be changed), transparent (viewable by authorized parties), and auditable (verifiable for accuracy).
- iii. **NIN Integration:** Creating an authentication layer that communicates with the NIN database (or simulated version) to verify voter identity and prevent multiple voting or impersonation.
- iv. **Performance Testing:** Evaluating the system in a simulated environment to measure transaction processing speed (performance), attack resistance (security), and system uptime (reliability).

Out of scope:

- i. Nationwide deployment across all polling units.
- ii. Legal, political, and policy adoption processes.
- iii. Physical hardware procurement, maintenance, or on-site election logistics.

1.8 Expected Outcome

The expected outcome is a fully functional blockchain-based e-voting system where eligible voters authenticate using their National Identification Number (NIN), securely cast votes, and have votes recorded immutably on a blockchain network. The system will provide secure voter verification, transparent vote storage, and automated vote tallying through smart contracts. A comprehensive

project report documenting the system's design, architecture, implementation, and testing will be produced.

1.9 Organization of the Study

This research is organized into five chapters:

Chapter One: Introduction

Presents the background, problem statement, aim and objectives, significance, scope, and definition of terms. It establishes the foundation for understanding why a blockchain-based e-voting system integrated with NIN authentication is necessary.

Chapter Two: Literature Review

Reviews relevant theoretical and empirical literature, examining existing electronic voting systems, blockchain technologies, biometric authentication methods, and previous studies integrating NIN or similar national identity systems for secure voting.

Chapter Three: System Design and Methodology

Describes the research methodology and system architecture, outlining system requirements, flowcharts, UML diagrams, blockchain model (consensus mechanism, ledger structure), and NIN verification process. It details the tools and technologies used for development.

Chapter Four: Implementation and Testing

Presents the practical implementation including development environment, user interface design, backend logic, smart contract deployment, and NIN authentication integration. It includes functional and non-functional testing, performance evaluation, and security validation.

Chapter Five: Summary, Conclusion, and Recommendations

Summarizes the entire work and discusses how the system meets stated objectives. It provides conclusions on the effectiveness of the blockchain-based e-voting model and offers recommendations for future enhancements, policy adoption, and further research.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction to the Research Topic

Electoral integrity remains a critical challenge in developing nations, where vote manipulation, fraud, and lack of transparency undermine public confidence in democratic processes (Kshetri & Voas, 2018). Blockchain-based e-voting systems offer a solution by leveraging blockchain's key features decentralization, immutability, transparency, and cryptographic security to create tamper-resistant voting infrastructures that can be audited by multiple stakeholders (Hjálmarsson et al., 2018).

Integrating Nigeria's National Identification Number (NIN) adds a crucial layer of identity verification. This ensures "one person, one vote" by preventing duplicate registrations and voter impersonation (Yaga et al., 2019). This research combines blockchain technology for secure vote recording with NIN-based authentication to create a verifiable electoral platform.

The system addresses Nigeria's specific challenges, manual vote counting across dispersed polling units, electoral malpractices like ballot stuffing, and the need for secure yet inclusive voting mechanisms (Adeniji & Olaosebikan, 2020). By enabling real-time vote counting and verification, the system can enhance electoral credibility, reduce disputes, and restore citizen confidence that votes are accurately recorded without manipulation (Pawlak et al., 2018).

2.2 Historical Context of the Research Topic

Electronic voting systems have evolved through several distinct phases over five decades. Traditional paper-based voting relies on manual counting and physical ballot boxes, making it vulnerable to manipulation, human error, and logistical challenges (Gritzalis, 2002). The first generation of electronic voting emerged in the 1960s with direct recording electronic (DRE) machines using push buttons or touch screens. However, these systems lacked verifiable paper trails and raised concerns about result manipulation (Mercuri, 2002).

The 1990s introduced internet-based voting systems. Estonia implemented remote e-voting for national elections, but these centralized systems remained vulnerable to server attacks and lacked transparency (Springall et al., 2014). The breakthrough came in 2008 when Bitcoin introduced

blockchain technology, demonstrating how distributed ledgers could create tamper-resistant records without central authority (Nakamoto, 2008). Researchers began exploring blockchain for voting in the early 2010s, recognizing its potential to address electoral integrity challenges (Ayed, 2017).

The first blockchain voting trials occurred between 2016-2018 in Switzerland, South Korea, and Sierra Leone, though these implementations faced scalability and usability challenges (Hjálmarsson et al., 2018). In Nigeria, the National Identity Management Commission (NIMC) launched the NIN system in 2014, creating the infrastructure necessary for digital identity verification in e-voting (NIPC, 2020). Recent research focuses on integrating biometric authentication with blockchain voting to strengthen identity verification while maintaining ballot secrecy (Alam et al., 2021). Challenges around voter privacy, accessibility, and regulatory frameworks continue to drive ongoing research.

2.2.1 Theoretical Frameworks

This research applies several established frameworks from information security and democratic systems. The CIA Triad (Confidentiality, Integrity, Availability) provides foundational security principles (Stallings & Brown, 2018). Confidentiality ensures votes remain secret and cannot be linked to voters. Integrity guarantees votes cannot be altered after submission. Availability ensures eligible voters can access the system during voting periods.

The Defense-in-Depth model informs the multi-layered security approach. Blockchain immutability, cryptographic encryption, NIN authentication, and smart contract validation function as complementary layers protecting against different attack vectors (Scarfone & Mell, 2007). The Zero Trust Security model requires continuous verification throughout the voting process rather than relying only on initial login (Rose et al., 2020). Voters authenticate using NIN before accessing ballots, votes are cryptographically signed to prevent tampering, and blockchain nodes independently verify each transaction. These frameworks guide the system architecture towards comprehensive security while maintaining ballot secrecy and electoral transparency.

2.3 Key Concepts and Definitions

- i. **Blockchain:** is a distributed ledger technology that records transactions across multiple computers, making records immutable and transparent. Each block contains a cryptographic hash of the previous block, creating a tamper-evident chain (Yaga et al., 2019).
- ii. **Smart Contracts:** are self-executing programs stored on a blockchain that automatically enforce predefined rules and conditions, enabling automated vote validation and tallying without human intervention (Zheng et al., 2020).
- iii. **Decentralization:** describes a system architecture where control and decision-making are distributed across multiple nodes rather than concentrated in a single central authority, reducing single points of failure and manipulation risk (Buterin, 2014).
- iv. **Immutability:** means that once data is recorded on a blockchain, it cannot be altered or deleted without detection, ensuring votes remain permanent and tamper-proof after submission (Nakamoto, 2008).
- v. **Consensus Mechanism:** refers to the algorithm blockchain networks use to achieve agreement among distributed nodes on transaction validity. Common mechanisms include Proof of Work, Proof of Stake, and Practical Byzantine Fault Tolerance, is a system that allows a network to reach a correct agreement even if up to one-third of its members are lying or failing, by having all nodes cross-verify each other's messages in multiple rounds. (Mingxiao et al., 2017).
- vi. **National Identification Number (NIN):** is a unique 11-digit number issued by Nigeria's National Identity Management Commission to citizens and legal residents, serving as the primary identifier for government services and identity verification (NIMC, 2020).
- vii. **Cryptographic Hashing:** uses mathematical algorithms to convert data into fixed-length strings that uniquely represent the original data, enabling vote verification without revealing vote content (Stallings & Brown, 2018).
- viii. **Digital Signature:** is a cryptographic technique that authenticates the sender of a digital message and ensures message integrity, allowing voters to sign their ballots without revealing their identity (Katz & Lindell, 2014).

- ix. **Electoral Integrity:** refers to the degree to which elections are conducted fairly, transparently, and according to democratic principles, producing results that accurately reflect voter preferences (Norris, 2014).
- x. **Voter Authentication:** involves verifying that a person attempting to vote is eligible and authorized to participate in the election, typically through identity verification mechanisms (Jones & Simons, 2012).
- xi. **Ballot Secrecy:** ensures that individual votes cannot be traced back to specific voters, protecting voters from coercion and enabling free expression of political preferences (Benaloh, 2006).
- xii. **End-to-End Verifiability:** means that voters can verify their votes were correctly recorded, counted, and included in the final tally without compromising ballot secrecy (Adida, 2008).

2.4 Methodologies and Research Designs

Research in blockchain-based e-voting systems primarily uses experimental, design science, and case study methodologies. Experimental approaches test blockchain voting platforms to measure transaction throughput, latency, scalability, and security resilience against attack scenarios (Hjálmarsson et al., 2018). These studies compare blockchain implementations against traditional systems or evaluate different consensus mechanisms for optimal configurations.

Design science research focuses on creating and evaluating new blockchain voting architectures for real-world electoral contexts (Kshetri & Voas, 2018). This involves iterative cycles of design, implementation, testing, and refinement based on stakeholder feedback. Case study methodologies examine actual deployments in Estonia, Switzerland, and Sierra Leone, documenting implementation challenges, voter acceptance rates, and operational outcomes (Hao et al., 2020). Mixed-methods approaches combine quantitative performance metrics with qualitative assessments of voter experiences and regulatory compliance (Pawlak et al., 2018).

Data collection methods vary across studies. Some researchers use simulated elections with controlled populations, while others analyze data from pilot deployments or actual electoral events. Proof of Authority (PoA) and Practical Byzantine Fault Tolerance (PBFT) consensus mechanisms are popular choices for voting applications, offering transaction confirmation times under 10 seconds compared to Proof of Work (Shahzad & Crowcroft, 2019). Ethereum and Hyperledger

Fabric dominate platform choices due to their mature development tools and extensive documentation.

Evaluation frameworks consider technical performance, security, voter privacy, auditability, accessibility, and regulatory compliance (Springall et al., 2014). Security testing includes penetration testing, formal verification of smart contracts, and simulation of attack scenarios such as double-voting attempts, denial-of-service attacks, and node collusion.

2.5 Empirical Studies

Recent research demonstrates both the potential and challenges of blockchain-based e-voting systems. Bismark et al. (2025) pioneered NIN-blockchain integration in Nigeria, achieving 100% authentication accuracy by separating identity verification from vote recording.

Njoku et al. (2023) used IBM Hyperledger Fabric with NIN authentication, showing that enterprise blockchains offer faster speeds and better privacy control than public blockchains. However, both studies tested only simulated systems without connecting to actual NIMC databases, leaving questions about real-world performance, network delays, and data reliability unanswered.

Jayakumari et al. (2024) tackled scalability by combining cloud computing with blockchain technology, achieving fast transaction times through automated smart contracts. Their cloud-based approach raised concerns about data sovereignty and reliance on third-party providers.

Osayomore et al. (2025) strengthened authentication by adding fingerprint and facial recognition to blockchain voting, demonstrating higher accuracy than NIN-only systems. However, the high costs of biometric hardware and privacy risks of centralized biometric databases pose significant barriers to nationwide deployment.

Steven et al. (2025) explored the Internet Computer Protocol in Indonesia, achieving over 1,000 transactions per second at lower costs than Ethereum. Their real-world implementation revealed important challenges including voter education needs and varying technical literacy across age groups and locations. While valuable, their Indonesian-specific design lacks NIN integration and does not address Nigerian legal requirements, limiting direct application to Nigerian elections.

2.6 Conceptual Frameworks

The conceptual framework for blockchain-based e-voting systems with NIN integration comprises interconnected layers. The identity layer manages voter registration, NIN verification, and biometric authentication to ensure only eligible citizens participate (Kshetri & Voas, 2018). This layer interfaces with Nigeria's National Identity Management Commission database to validate credentials before granting access.

The blockchain layer provides distributed ledger infrastructure where votes are recorded as immutable transactions. Each vote is cryptographically hashed and linked to previous transactions, creating a tamper-evident chain (Yaga et al., 2019). Consensus mechanisms allow network nodes to agree on vote validity without central authority.

The smart contract layer contains self-executing programs that enforce voting rules, validate eligibility, prevent double voting, and automatically tally results (Hjálmarsson et al., 2018). The security layer implements cryptographic encryption for confidentiality, digital signatures for authentication, and access controls separating voter identity from vote content to maintain ballot secrecy (Shahzad & Crowcroft, 2019).

The interface layer provides user-facing applications for voters, election officials, and observers, emphasizing simplicity for varying technical abilities (Risnanto & Imran, 2019). The verification layer enables end-to-end auditability, allowing voters to confirm votes were recorded correctly while preventing identification of how individuals voted (Adida, 2008).

This framework emphasizes modularity for independent component development and testing. Privacy-preserving design includes data minimization, ballot secrecy, and electoral law compliance. The model balances technological capabilities with democratic principles to achieve legitimacy (Norris, 2014).

2.7 Debates and Controversies

Significant debates surround the implementation of blockchain-based e-voting systems in democratic elections. Privacy concerns constitute the primary controversy. Critics argue that digital voting systems inherently create electronic trails that could link voters to their ballot choices, compromising the fundamental democratic principle of secret ballots (Springall et al., 2014). Opponents fear that government authorities or hackers could analyze voting patterns and

identify individual voters through transaction metadata, network analysis, or correlation with registration databases.

1. **Scalability challenges** present another contentious issue. Blockchain networks face throughput limitations that may prevent processing millions of votes within short timeframes during national elections (Shahzad & Crowcroft, 2019). Critics point to Ethereum's 15-30 transactions per second as insufficient for large populations, though proponents argue that permissioned blockchains and layer-2 solutions can achieve necessary performance levels.
2. **Technological accessibility** raises concerns that blockchain voting may disadvantage citizens lacking internet connectivity, smartphones, or technical literacy, potentially creating digital divides that exclude vulnerable populations (Khan et al., 2021). With researchers documenting exploitable weaknesses in smart contracts, client-side software, and network infrastructure that could enable vote manipulation despite blockchain's theoretical immutability (Atzei et al., 2017).
3. The **verifiability paradox** creates philosophical tensions. Systems providing strong voter verification may inadvertently enable vote selling or coercion by allowing voters to prove how they voted to third parties (Benaloh, 2006). Regulatory uncertainty presents practical challenges, as most national electoral laws were written for paper-based systems and do not clearly address blockchain voting legality, recount procedures, or dispute resolution (Pawlak et al., 2018).
4. **Trust in technology versus trust in institutions** remains fundamentally contested. Some argue that blockchain's cryptographic guarantees provide stronger assurances than human-administered elections, while others maintain that democratic legitimacy requires human oversight that algorithms cannot replace (Norris, 2014). Cost-benefit calculations generate ongoing disagreement, with implementation requiring substantial upfront investments while cost savings materialize only over extended periods (Khan et al., 2021).
5. **Environmental impact** concerns have emerged regarding energy consumption, particularly for Proof of Work blockchains. These controversies have prompted varied regulatory responses globally, with some jurisdictions embracing blockchain voting pilots while others impose strict prohibitions.

2.8 Theoretical Contributions

This research proposes an integrated framework combining blockchain technology with Nigeria's National Identification System to create a secure electoral architecture. The study demonstrates how distributed ledger immutability, cryptographic authentication, smart contracts, and biometric verification function as complementary security layers for democratic elections. By documenting design principles and trade-offs between transparency and privacy, the research provides insights applicable to broader governance contexts.

The work contributes to democratic theory by exploring how technology-mediated voting differs from traditional ballot casting and its implications for political participation and electoral legitimacy in Nigeria. It extends the CIA Triad (Confidentiality, Integrity, Availability) framework by demonstrating its application where these principles must coexist despite inherent tensions, advancing zero-trust security models requiring continuous cryptographic verification.

By examining NIN-based biometric authentication with blockchain anonymity, this research contributes to identity management theory specifically how systems verify identity during authentication while preventing identity linkage during vote recording. The work demonstrates how smart contracts encode Nigerian electoral rules into executable logic.

2.9 Methodological Contributions

Methodologically, this research demonstrates systematic blockchain-NIN database integration while maintaining data sovereignty and privacy compliance. The study contributes methods for implementing ballot secrecy in transparent blockchain systems through cryptographic techniques enabling vote verification without revealing content.

The research presents blockchain optimization techniques through consensus mechanism selection, transaction batching, and scalability strategies. The evaluation framework assesses technical performance, security, usability, and regulatory compliance offering a comprehensive template for blockchain governance systems. The study advances voter privacy protection by evaluating linkability risks, transaction metadata exposure, and anonymity guarantees. It demonstrates hybrid system design combining blockchain immutability with database efficiency, identifying when decentralization is necessary versus when centralized systems suffice.

2.10 Practical Implications

The practical implications of this research extend across multiple stakeholder groups in Nigeria's electoral ecosystem. For the Independent National Electoral Commission (INEC), the study demonstrates the feasibility of implementing a cost-effective blockchain-based voting system that can scale to handle millions of voters across Nigeria's 36 states and Federal Capital Territory without proportional increases in staffing or manual counting requirements. The research provides guidance for electoral bodies considering blockchain adoption, outlining technical requirements, infrastructure needs, and potential challenges specific to Nigerian contexts.

For Nigerian voters, the system offers transparent, verifiable voting that eliminates opportunities for ballot stuffing and result manipulation while ensuring each vote is accurately recorded and counted. Blockchain transparency may increase voter confidence compared to opaque manual counting processes that currently generate frequent disputes. For political parties and candidates, automated vote tallying and real-time result verification reduce opportunities for electoral fraud while providing credible audit trails that can resolve disputes without prolonged legal battles.

For the National Identity Management Commission, the research demonstrates practical applications of the NIN database beyond basic identification, showing how biometric records can strengthen democratic processes while justifying continued investment in national identity infrastructure. For civil society organizations and election observers, the system provides unprecedented transparency through blockchain's public ledger, enabling independent verification of results and strengthening democratic accountability.

For Nigerian policymakers and lawmakers, the research illuminates the capabilities and limitations of blockchain voting technology, informing potential amendments to electoral laws and regulatory frameworks that would enable legal adoption. The study's emphasis on privacy-preserving design and compliance with democratic norms provides practical insights for developing systems that balance technological efficiency with constitutional requirements for free, fair, and credible elections. For technology developers and system integrators, the research provides a practical blueprint for implementing blockchain voting solutions in developing country contexts where infrastructure constraints and diverse user populations require careful design considerations.

2.11 Related Works

A comprehensive review of empirical literature reveals the evolution and current state of blockchain-based e-voting systems integrated with national identity verification mechanisms.

Table 2.1 synthesizes major research contributions, highlighting their methodological approaches, key findings, and limitations. The studies are organized thematically based on core research focus: blockchain architecture, biometric integration, NIN-specific implementations, and security frameworks.

Table 2.1: Review of Related Works

S/N	NAME	TITLE	METHOD	FINDINGS	LIMITATIONS
1	Bismark et al. (2025)	Development of a blockchain-based electronic voting system utilizing NIN	Ethereum blockchain with smart contracts; NIN database verification; local testing	Achieved 100% login authentication; secure, transparent vote recording using NIN verification	Limited to local testing; nationwide scalability untested; no actual NIMC database integration
2	Njoku et al. (2023)	Development of an e-voting system using IBM blockchain technology and NIN	IBM Hyperledger Fabric; SSADM; NIN database integration	Web-based platform authenticates users via NIN before blockchain ballot access	Enterprise IBM framework limits cost efficiency; no exploration of gas-efficient blockchains
3	Jayakumari et al. (2024)	E-voting system using cloud-based hybrid blockchain technology	Hybrid blockchain; smart contracts; timestamp authentication; PBFT consensus	Secure, transparent, cost-effective voting; automated tallying via smart contracts	Simulation only; no real deployment; lacks national identity authentication
4	Dumbre et al. (2025)	Design and implementation of a blockchain-based e-voting system with enhanced voter authentication	Permissioned blockchain; PoA consensus; smart contracts; multi-factor authentication	Secure, tamper-proof voting with low latency; automated registration and tallying	Simulation only; real-world deployment untested; no national ID integration
5	Osayomore et al. (2025)	Blockchain-based e-voting system enhanced with	Blockchain with biometric authentication;	Improved authentication accuracy;	High infrastructure costs; biometric

		fingerprint and facial recognition	SPSS analysis; SWOT evaluation	enhanced trust; demonstrated vote immutability	privacy concerns; uncertain nationwide scalability
6	Ejenarhome et al. (2025)	Evaluation of electronic-based voting system and design (2024 Edo State election)	Blockchain with multimodal biometrics; case study of electoral failures	Decentralized ledgers prevent vote alteration; solution to ballot snatching	Heavy focus on physical hardware; limited NIMC software integration details; scalability unaddressed
7	Steven et al. (2025)	Implementation of blockchain-based electronic voting system: Indonesia case study	Internet Computer Protocol; Azle (TypeScript); facial recognition	Faster, cheaper blockchain using ICP; prevented vote manipulation via Face ID	Designed for Indonesian context; no Nigerian NIMC API integration
8	Pandey & Tiwari (2023)	Secure e-voting systems using blockchain technology	Ethereum; smart contracts; Ganache; MetaMask; Web3.js; Truffle; Mocha	Secure, transparent, anonymous voting; improved on-chain verification	High Ethereum gas costs; not tested in live elections; requires technical expertise
9	Russo et al. (2021)	Chirotonia: A scalable and secure e-voting framework based on blockchains and linkable ring signatures	Blockchain with linkable ring signatures; privacy-preserving cryptography	Voter anonymity preserved while preventing double voting; scalable architecture	Complex cryptographic implementation; limited real-world testing; unclear high-load performance
10	Shahzad & Crowcroft (2019)	Trustworthy electronic voting using adjusted blockchain technology	Modified blockchain; PBFT consensus; transaction optimization	Transaction confirmation under 10 seconds; trustworthy Byzantine fault-tolerant recording	Permissioned blockchain limits transparency; centralized nodes contradict decentralization
11	Hjálmarsson et al. (2018)	Blockchain-based e-voting system	Ethereum; university election pilot (50 participants)	Demonstrated technical feasibility; validated vote casting and tallying	Small-scale only (50 voters); national scalability concerns; throughput limitations

12	Khan & Salah (2021)	IoT security: Review, blockchain solutions, and open challenges	Literature review; blockchain security framework analysis	Blockchain enhances data immutability, device authentication, distributed trust in IoT	Conceptual only; no implementation; scalability and resource demands remain challenges
13	Hardwick et al. (2018)	E-voting with blockchain: An e-voting protocol with decentralization and voter privacy	Hyperledger Fabric comparison; privacy protocols; performance benchmarking	Hyperledger 5-7x faster than Ethereum; 90% lower energy consumption	Permissioned blockchain limits public audit; complex setup for authorities
14	Kshetri & Voas (2018)	Blockchain-enabled e-voting	Theoretical analysis; security assessment framework	Blockchain addresses vote tampering, enables transparent auditing, reduces fraud	Theoretical only; practical deployment challenges unaddressed; regulatory barriers unexplored
15	Yavuz et al. (2018)	Towards secure e-voting using Ethereum blockchain	Ethereum smart contracts; homomorphic encryption; prototype	Vote tallying without decrypting individual ballots; preserved voter privacy	High computational overhead; processing time unsuitable for time-sensitive elections

CHAPTER THREE

SYSTEM DESIGN AND METHODOLOGY

3.1 Research Approach

This research employs a mixed-methods approach, combining quantitative and qualitative elements to address the design, implementation, and evaluation of the blockchain-based e-voting system integrated with National Identification Number (NIN) authentication. The quantitative component measures system performance through metrics such as transaction throughput (votes processed per second), authentication accuracy rates, blockchain synchronization latency, smart contract execution time, and system scalability under varying voter loads. Performance benchmarking compares the implemented system against existing electronic voting solutions in terms of vote recording speed, result tallying efficiency, and infrastructure resource consumption.

The qualitative component includes evaluation of user experience through interface usability assessments, voter confidence surveys, and system functionality validation with stakeholders including potential voters, election officials, and technical administrators. User acceptance testing gathers feedback on interface clarity, authentication process simplicity, voting workflow intuitiveness, and overall trust in the system's ability to record votes accurately and securely.

This mixed-methods approach provides a complete understanding of both the technical performance capabilities and practical usability of the developed system, ensuring that the solution addresses Nigeria's electoral integrity challenges while remaining accessible and trustworthy for end users across diverse demographic groups and technical literacy levels.

3.2 Research Design

This research uses experimental design and design science methodology. The study designs, implements, and tests the blockchain e-voting system to evaluate its effectiveness. The research has four stages: architecture design, prototype development, functional testing, and performance evaluation.

The architecture design phase selects the blockchain platform, determines the consensus mechanism, specifies smart contract logic, and establishes NIN integration protocols. The prototype development phase builds core components incrementally: voter registration module,

NIN authentication interface, blockchain vote recording mechanism, and smart contract-based tallying.

Functional testing simulates electoral processes including voter registration, authentication attempts (legitimate and fraudulent), vote casting, and result tallying. This verifies the system enforces electoral rules and prevents unauthorized actions. Performance evaluation measures authentication accuracy, transaction processing speed, scalability under voter loads, and blockchain synchronization efficiency. Usability testing with representative voter demographics assesses user experience.

This design validates whether the system addresses Nigeria's electoral integrity challenges while remaining practical and accessible for real-world deployment.

3.3 Software Development Methodology

This project adopts an adapted Agile methodology, specifically Scrum, due to the iterative nature of blockchain smart contract development, NIN integration testing, and the need for continuous security validation and refinement.

3.3.1 Adopted Methodology

An adapted Agile (Scrum) methodology was selected for this project based on several project-specific factors:

- i. **Iterative Blockchain Development:** Smart contracts for vote recording, authentication, and tallying require continuous refinement. Agile's sprint-based approach enables iterative improvement of security, efficiency, and compliance.
- ii. **Evolving Security Requirements:** Security vulnerabilities discovered during development require requirement updates. Agile accommodates these changes without complete project restructuring.
- iii. **Early Component Testing:** Each sprint delivers functional modules (NIN authentication, blockchain vote recording, smart contract execution). This enables early testing before full integration, allowing rapid issue identification and correction.
- iv. **Complexity Management:** The system is broken into manageable features across sprints: voter registration, authentication interface, vote casting, and result tallying. This reduces complexity and enables focused development.

- v. **Academic Timeline Constraints:** This final year project has strict submission deadlines. Agile sprints ensure consistent progress within the required timeframe.

The Scrum implementation uses 2-week sprint cycles. Each sprint focuses on specific system components. Sprint planning defines objectives and deliverables. Daily tracking monitors progress. Sprint reviews evaluate outputs against acceptance criteria. Sprint retrospectives identify process improvements.

3.4 Methodology: Tools and System Architecture

1. **Development Stack** The system is built on the Ethereum ecosystem. Solidity is used for smart contract logic, while the Truffle Suite and Ganache provide the framework for development, testing, and local blockchain emulation. For the interface, React.js (Frontend) and Node.js/Express (Backend) handle the voter experience and API requests, respectively.
2. **Authentication and Security** To verify identity without compromising privacy, the system integrates the WebAuthn API. This allows for device-level biometric authentication (fingerprint or facial recognition) where sensitive data never leaves the user's device. For data integrity, AES-256 encrypts voter information, while Elliptic Curve Cryptography (ECC) ensures secure digital signatures for every cast vote.

3.4.1 Core Technical Tools

- i. **Blockchain:** Solidity, Web3.js, OpenZeppelin (Security libraries).
- ii. **Database:** MongoDB for registration logs; a simulated **NIN Database** for NIMC verification.
- iii. **Security:** JWT for session management; bcrypt for hashing credentials.
- iv. **Testing:** Mocha/Chai for smart contracts; Postman for API validation.

3.4.2 System Architecture and Workflow

The integration of NIN and Blockchain follows a four-stage process to ensure "one person, one vote."

1. **NIN Verification:** The voter enters their 11-digit NIN. The system queries the simulated NIMC database to confirm eligibility and checks the blockchain ledger to ensure the user hasn't already registered.
2. **Biometric Enrollment (WebAuthn):** First-time users link their NIN to their device's biometrics. WebAuthn generates a cryptographic key pair; the biometric data stays on the device, while the public key is stored for future authentication.
3. **Authentication & Vote Casting:** For subsequent logins, the system issues a biometric challenge. Upon a successful scan, the voter accesses the ballot. The selected vote is encrypted and sent as a blockchain transaction.
4. **Immutability & Double-Voting Prevention:** The smart contract maps a hashed version of the NIN to a "voted" status. Once a vote is recorded on the distributed ledger, the blockchain's immutability prevents any alteration, and the system automatically rejects further attempts from that NIN.

3.5 Research Timeline and Evaluation

The research spans approximately 16 weeks, divided into distinct phases as shown in Table 3.1:

Table 3.1: Research Timeline and Activities

Phase	Week(s)	Activities
Planning and Requirements Analysis	1-2	Literature review completion System requirements specification (functional and non-functional) Methodology finalization Development environment setup (Ganache, Truffle/Hardhat, Node.js, React) NIN database simulation design

Core System Development (Sprint 1 & 2)	3-6	Voter registration module implementation NIN verification interface development WebAuthn biometric authentication integration Database schema design and implementation Basic voting interface creation
Blockchain Integration (Sprint 3 & 4)	7-10	Smart contract development for vote recording Smart contract logic for duplicate voting prevention Blockchain connection implementation using Web3.js Vote encryption and submission mechanism Result tallying smart contract development
System Integration and Testing (Sprint 5)	11-13	Component integration into unified system Authentication flow testing (NIN + WebAuthn) Vote recording and retrieval testing Duplicate voting prevention validation Comprehensive system testing with simulated electoral scenarios
Evaluation and Refinement	14-15	Performance metrics collection and analysis Usability testing with representative users Security testing (authentication bypass attempts, duplicate voting tests) Bug fixes and system refinements Interface improvements based on user feedback
Documentation and Reporting	16	Final documentation completion Research report writing (Chapters 4 and 5) System demonstration preparation Presentation materials creation

3.6 Data Analysis and Evaluation Plan

Performance Metrics

The system's effectiveness is measured through four primary indicators:

1. **Authentication Accuracy:** The percentage of legitimate NIN holders correctly verified via WebAuthn versus rejected unauthorized attempts.
2. **Transaction Speed:** The time elapsed from vote submission to blockchain confirmation.
3. **Security Integrity:** The rate at which the smart contract successfully blocks duplicate voting attempts.
4. **System Throughput:** The capacity to handle concurrent voters without performance lag.

Evaluation Method

The system is evaluated using controlled test cases with predetermined outcomes to verify technical and security requirements.

Test Case Categories

1. **Legitimate Scenarios:** Verifies NIN-biometric registration, concurrent voting sessions, and post-vote verification.
2. **Security Violations:** Tests the system's ability to reject invalid NINs, block duplicate votes via smart contracts, and deny access upon biometric failure.
3. **Edge Cases:** Assesses system resilience during network interruptions, biometric timeouts, and cross-browser compatibility.

3.7 Validity and Reliability

To ensure research integrity, the following measures are implemented:

1. **Internal Validity:** Controlled scenarios with predefined inputs and outputs eliminate external variables. Using a simulated NIN database ensures that system behaviors reflect design logic rather than external network fluctuations.

2. **External Validity:** Testing covers realistic electoral conditions, including varying voter loads (single vs. peak concurrent users), diverse devices (mobile vs. desktop), and different network latencies to ensure findings apply to real-world deployment.
3. **Reliability:** The deterministic nature of Smart Contracts ensures that identical inputs always produce identical outputs. Repeated testing of authentication workflows confirms consistent acceptance or rejection of credentials.
4. **Measurement & Construct Validity:** Metrics like Authentication Accuracy and Transaction Time are benchmarked against existing blockchain literature. These criteria accurately measure the system's ability to enforce electoral rules and maintain responsiveness.

3.8 System Design and Methodology for Software Development

3.8.1 System Architecture

The system follows a multi-tier architecture with four layers:

1. Presentation Layer (Frontend)

1. Provides user-facing interfaces accessible through web browsers:
2. Voter Registration: Captures NIN, personal details, initiates WebAuthn biometric enrollment
3. Authentication: Validates identity through NIN verification and WebAuthn biometric challenge
4. Voting: Displays candidates and ballot options for selection
5. Vote Confirmation: Provides cryptographic receipt with transaction hash
6. Result Verification: Enables verification of recorded votes on blockchain

The frontend uses responsive design for accessibility across devices. WebAuthn leverages browser's native biometric capabilities (fingerprint, facial recognition) without transmitting biometric data to servers.

2. Application Layer (Backend)

Contains core business logic:

1. Voter Registration Management: Processes registration, validates NIN against simulated NIMC database
2. Authentication Service: Handles NIN verification, manages WebAuthn challenges
3. Vote Processing: Receives votes, encrypts ballot data, prepares blockchain transactions
4. Blockchain Interaction: Communicates with Ganache using Web3.js, submits transactions
5. Result Tallying: Queries smart contracts for vote counts, generates reports

3. Blockchain Layer

Handles decentralized vote recording:

1. Voter Registry Contract: Maps registered voters (hashed NIN $\rightarrow$ status), prevents duplicates
2. Vote Recording Contract: Accepts encrypted votes, validates eligibility, records immutably
3. Duplicate Prevention Contract: Checks if voter has voted, rejects subsequent attempts
4. Result Tallying Contract: Counts votes per candidate, provides transparent totals
5. Consensus: Ganache uses instant mining, simulating Ethereum's proof-of-authority.

4. Data Layer

Manages off-chain storage:

1. Voter Database: Stores voter profiles with hashed NIN, timestamps, WebAuthn credential IDs
2. Authentication Logs: Records login attempts with timestamps, status, IP addresses
3. System Configuration: Maintains election parameters (start/end times, candidates, rules)
4. Session Management: Temporary storage for authenticated sessions using JWT

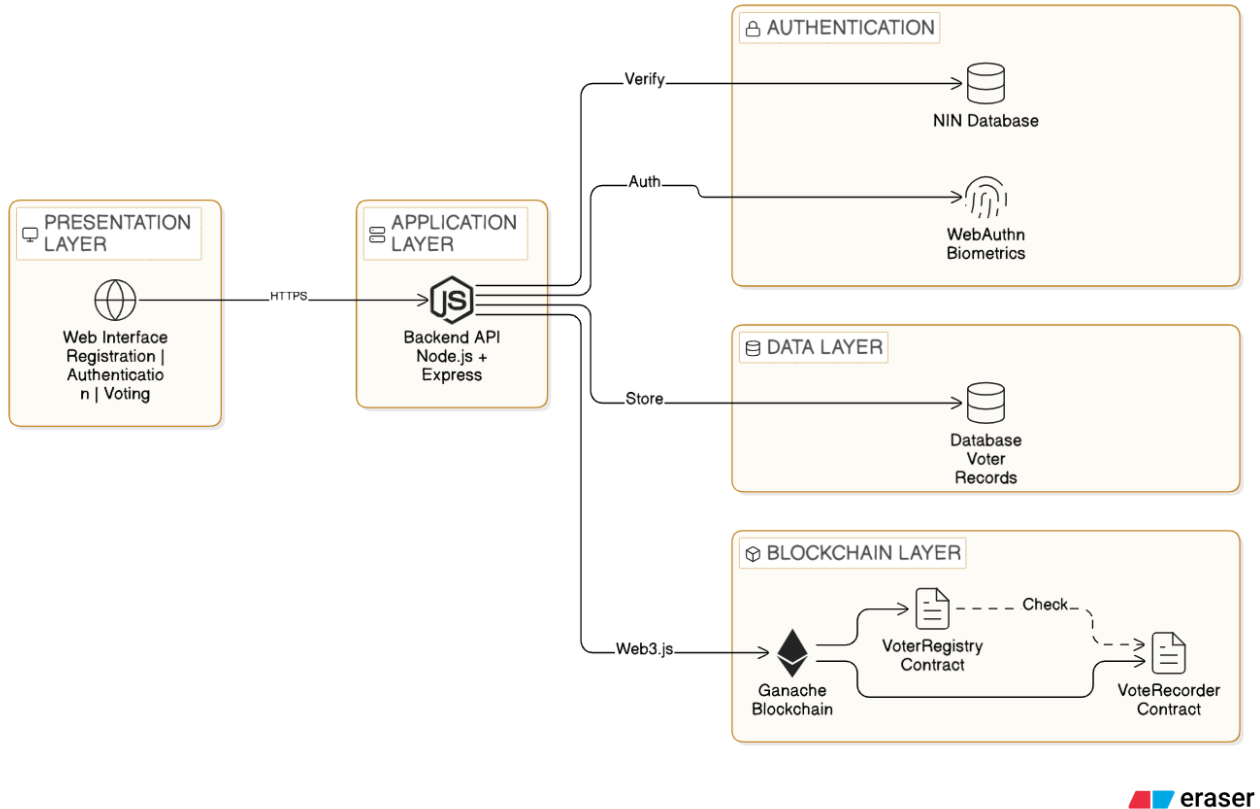


Figure 3.1: System Architecture Diagram

3.8.2 System Workflow

Phase 1 - Voter Registration:

1. Voter enters 11-digit NIN and personal information
2. Backend verifies NIN validity against simulated NIMC database
3. System checks blockchain to ensure NIN hasn't been registered
4. If valid, voter proceeds to WebAuthn biometric enrollment
5. Browser prompts fingerprint or facial biometric registration
6. WebAuthn generates public/private key pair; public key stored in database, private key stays on device
7. Voter record created and Voter Registry contract updated

Phase 2 - Voter Authentication:

1. Voter enters NIN on voting page
2. Backend retrieves WebAuthn credential ID for that NIN
3. System initiates WebAuthn authentication challenge
4. Browser prompts biometric verification (fingerprint/face scan)
5. Device validates biometric locally (data never leaves device)
6. If match, device signs challenge with private key
7. Backend verifies signature using stored public key
8. Successful authentication generates JWT session token.

Phase 3 - Vote Casting:

1. Voter views, selects candidate and clicks "Submit Vote"
2. Frontend encrypts vote and packages as blockchain transaction
3. Backend calls Vote Recording smart contract using Web3.js
4. Smart contract validates: voter registered, hasn't voted, election active
5. If valid, contract records vote and updates status to "has voted" then Blockchain returns transaction hash
6. Frontend displays success message with transaction hash

Phase 4 - Duplicate Voting Prevention:

1. Previously voted voter attempts to vote again and authenticates successfully
2. Smart contract executes duplicate prevention logic
3. Contract finds "has voted" flag set to true
4. Transaction rejected with error "Voter has already cast ballot"
5. Frontend displays rejection notification
6. Original vote remains unchanged (immutability preserved)

Phase 5 - Result Tallying:

1. After election closes, Result Tallying contract is triggered
2. Contract counts totals for each candidate.
3. Final tallies stored on blockchain, publicly accessible.
4. Any stakeholder can verify results by querying blockchain.
5. Vote immutability ensures results cannot be altered.



Figure 3.2: Complete Workflow

3.8.3 System Modelling

The system functionality is modelled using Unified Modelling Language (UML) diagrams:

1. Use Case Diagram

Actors:

- Voter: Registers, authenticates, and casts votes
- Election Administrator: Manages elections, monitors voting, and accesses results
- Election Observer: Verifies electoral integrity and result accuracy

Use Cases:

Voter:

1. Register as Voter (includes NIN Verification and WebAuthn Enrollment)
2. Authenticate Identity (includes NIN and Biometric Verification)
3. Cast Vote (requires Authentication)
4. Verify Vote Recorded

Administrator:

1. Configure Election (candidates, dates, rules)
2. Monitor Voting Activity
3. Access Results
4. Generate Reports

Observer:

1. Verify Blockchain Integrity

2. Audit Vote Records

3. Download Electoral Data

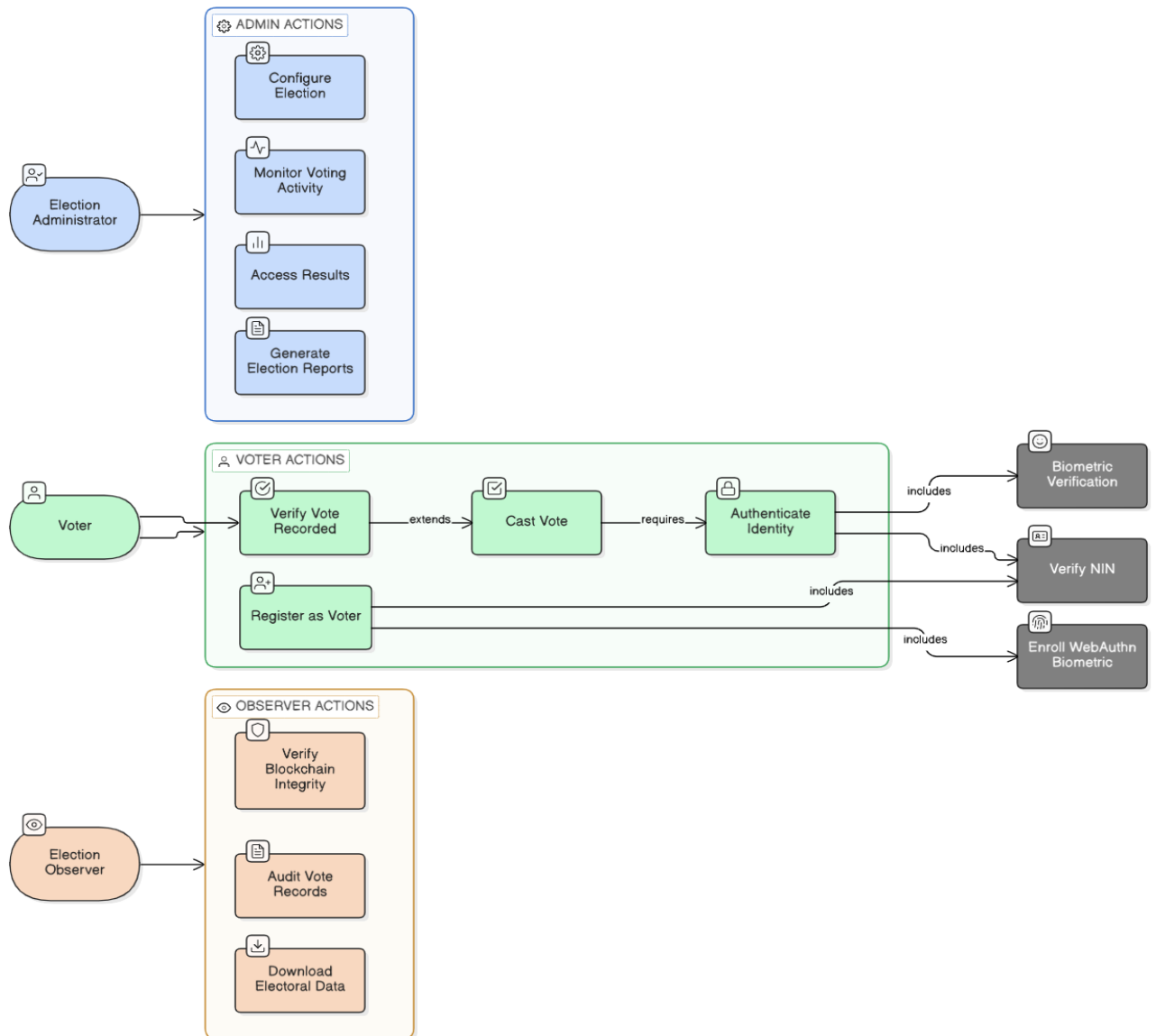


Figure 3.3: Use Case Diagram

2. Activity Diagram - Voting Workflow

Start → Voter navigates to voting website

Decision: Is voter registered?

If No → Registration Flow:

- Enter NIN and personal details
- System validates NIN with NIMC database
- Decision: NIN valid?
 - No → Display error → End
 - Yes → Check blockchain for existing NIN
 - Decision: Already registered?
 - Yes → Display error → End
 - No → Browser prompts biometric registration → User scans fingerprint/face → System stores WebAuthn public key → Registration complete → Proceed to Authentication

If Yes → Authentication Flow:

- Voter enters NIN
- System retrieves WebAuthn credential
- Browser prompts biometric verification → User scans fingerprint/face
- Decision: Biometric verified?
 - No → Display error → End
 - Yes → System generates session token → Proceed to Voting

Voting Flow:

- Display ballot with candidate list

- Voter selects candidate and submits vote
- System encrypts vote and creates blockchain transaction
- Smart contract validates: voter registered, hasn't voted, election active
- Decision: Validations passed?
 - No → Display error (reason: duplicate vote/election closed/not registered) → End
 - Yes → Smart contract records vote → Update voter status to "voted" → Generate transaction hash → Display success with hash → End

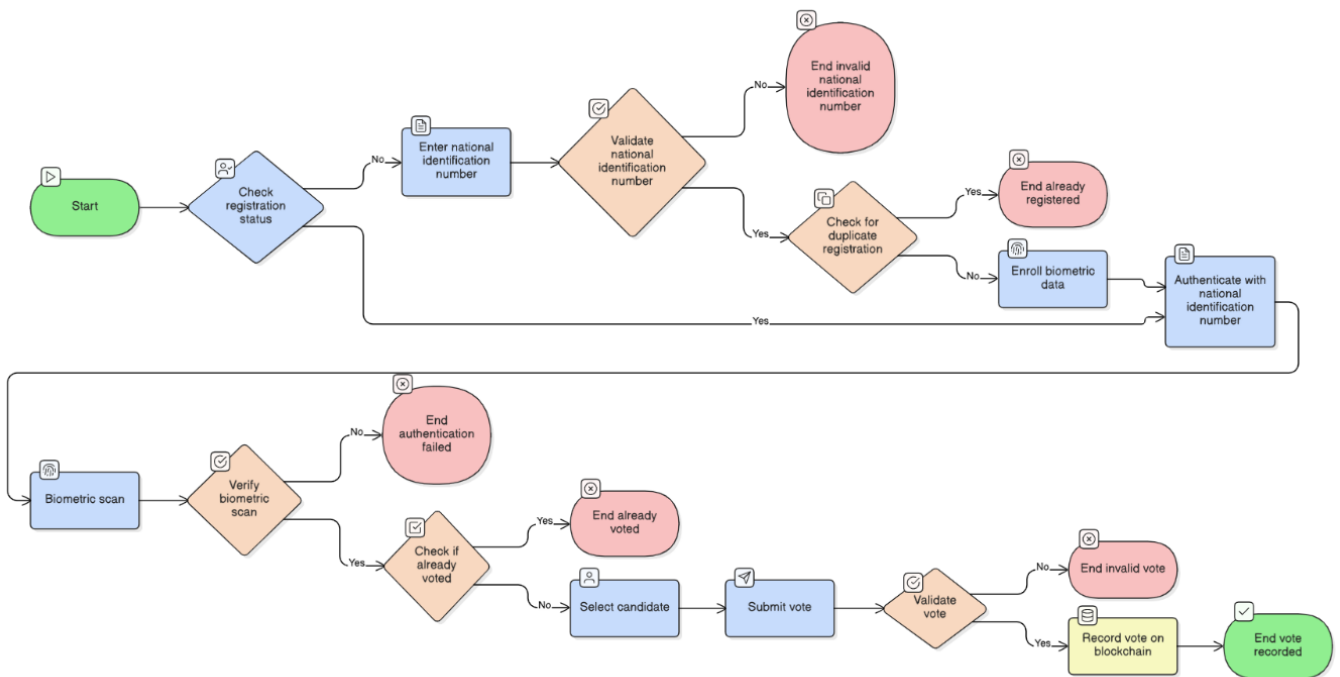


Figure 3.4: Activity Diagram - Voting Process/ Workflow

3. Sequence Diagram - Authentication Process

Objects: Voter, Web Browser, API Server, Voter Database, WebAuthn Service, Blockchain

Interaction Sequence:

1. Voter → Browser: Enters NIN on login page
2. Browser → API Server: POST /api/auth/login {nin: "12345678901"}
3. API Server → Voter Database: SELECT * FROM voters WHERE nin_hash = hash("12345678901")
4. Voter Database → API Server: Returns {voter_id, webauthn_credential_id, voting_status}
5. API Server → Browser: Returns {credential_id, challenge: "random_bytes"}
6. Browser → WebAuthn Service: navigator.credentials.get({challenge, credentialId})
7. WebAuthn Service → Voter: Prompts biometric verification (fingerprint/face scan)
8. Voter → WebAuthn Service: Provides biometric on device
9. WebAuthn Service: Validates biometric locally
 - **If No:** Returns error "Biometric verification failed" → Browser displays "Authentication failed" → End
 - **If Yes:** Returns signed authentication response → Continue
10. Browser → API Server: POST /api/auth/verify {credential_response, signature}
11. API Server: Verifies signature using stored public key
 - **If Invalid:** Returns error → Browser displays "Authentication failed" → End
 - **If Valid:** Continue
12. API Server → Blockchain: Query Voter Registry: hasVoted(voter_id)
13. Blockchain → API Server: Returns voting_status
14. API Server: Generates JWT session token {voter_id, exp: 3600}

15. API Server → Browser: Returns {token, voting_status, message: "Authentication successful"}

16. Browser → Voter: Redirects to ballot page (if not voted) or displays "Already voted"

End Sequence

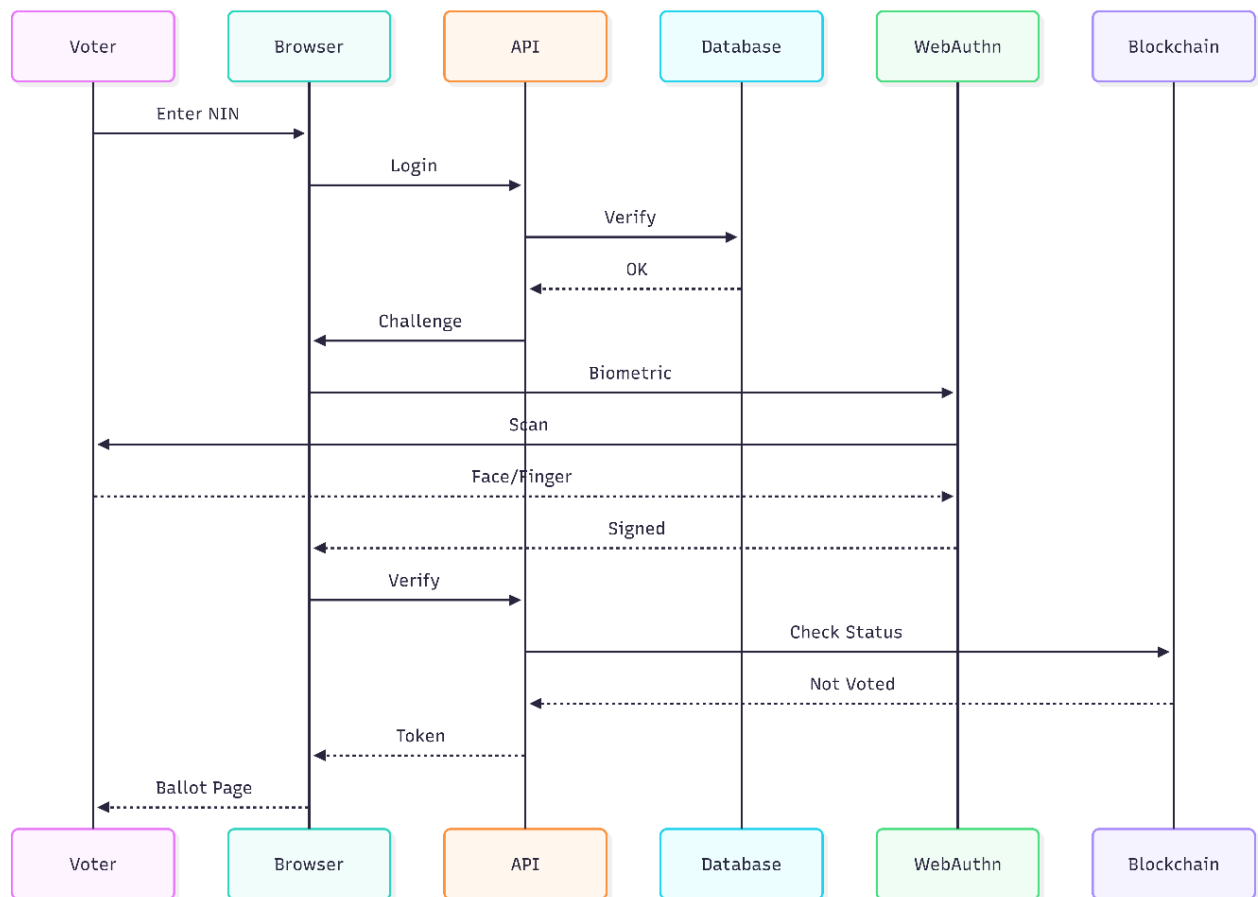


Figure 3.5: Sequence Diagram - Authentication Process

3.8.4 System Requirements

System requirements are categorized into functional and non-functional requirements, prioritized using MoSCoW method (Must have, Should have, Could have, Won't have).

Table 3.2: Functional Requirements

Priority	ID	Category	Description
Must Have	FR1- FR3	Registration & Authentication	Validate voter NIN against simulated NIMC database; implement WebAuthn biometric authentication; prevent duplicate registration
	FR4- FR6	Vote Recording	Record votes immutably on blockchain; prevent duplicate voting; validate votes only during active election period
	FR7- FR9	Security & Tallying	Encrypt vote data before blockchain submission; generate unique transaction hashes as receipts; automatically tally votes
	FR10- FR11	User Feedback	Provide real-time vote confirmation; reject unauthorized attempts with clear error messages
Should Have	FR12- FR14	Administration	Administrator dashboard for voting statistics; detailed audit logs; enable observers to verify counts by querying blockchain
	FR15- FR17	User Experience	User-friendly error messages; voter verification page for transaction hash checking; export results in standard formats (PDF, CSV)
Could Have	FR18- FR21	Enhanced Features	Multiple simultaneous elections; multi-language support (English, Hausa, Yoruba, Igbo); SMS/email notifications; graduated security responses
Won't Have	FR22- FR25	Excluded Features	No actual NIMC integration (simulation only); no vote modification after submission; no ranked-choice voting; no biometric quality assessment

Table 3.3: Non-Functional Requirements

Priority	ID	Category	Description
Must Have	NFR1-NFR2	Performance	Process votes with blockchain confirmation under 15 seconds; maintain 99% uptime during election
	NFR3	Security	Encrypt NIN using SHA-256 hashing; implement HTTPS/TLS; store WebAuthn credentials on user devices only
	NFR4-NFR6	Accessibility	Support standard browsers without specialized software; function on minimum specs (4GB RAM, dual-core processor)
	NFR8	Design	Implement responsive design for desktop and mobile
Should Have	NFR9-NFR11	User Experience	Achieve 95%+ authentication accuracy; handle 50+ concurrent users; provide intuitive interfaces
	NFR12-NFR14	Standards	Maintain error logging; implement session timeouts (15-30 minutes); comply with WCAG 2.1 Level A
Could Have	NFR15-NFR18	Enhanced Performance	Transaction processing under 5 seconds; graceful degradation on slower networks; PWA features; performance dashboards
Won't Have	NFR19-NFR22	Excluded Features	No Internet Explorer support; no advanced load balancing; no automated disaster recovery; no third-party penetration testing

3.8.5 Database Schema Design

1. Entity-Relationship Diagram (ERD) Database Relationships:

The database implements one-to-many relationships; Elections contain multiple Candidates; Admins create multiple Elections, and Voters generate multiple Authentication_Logs entries.

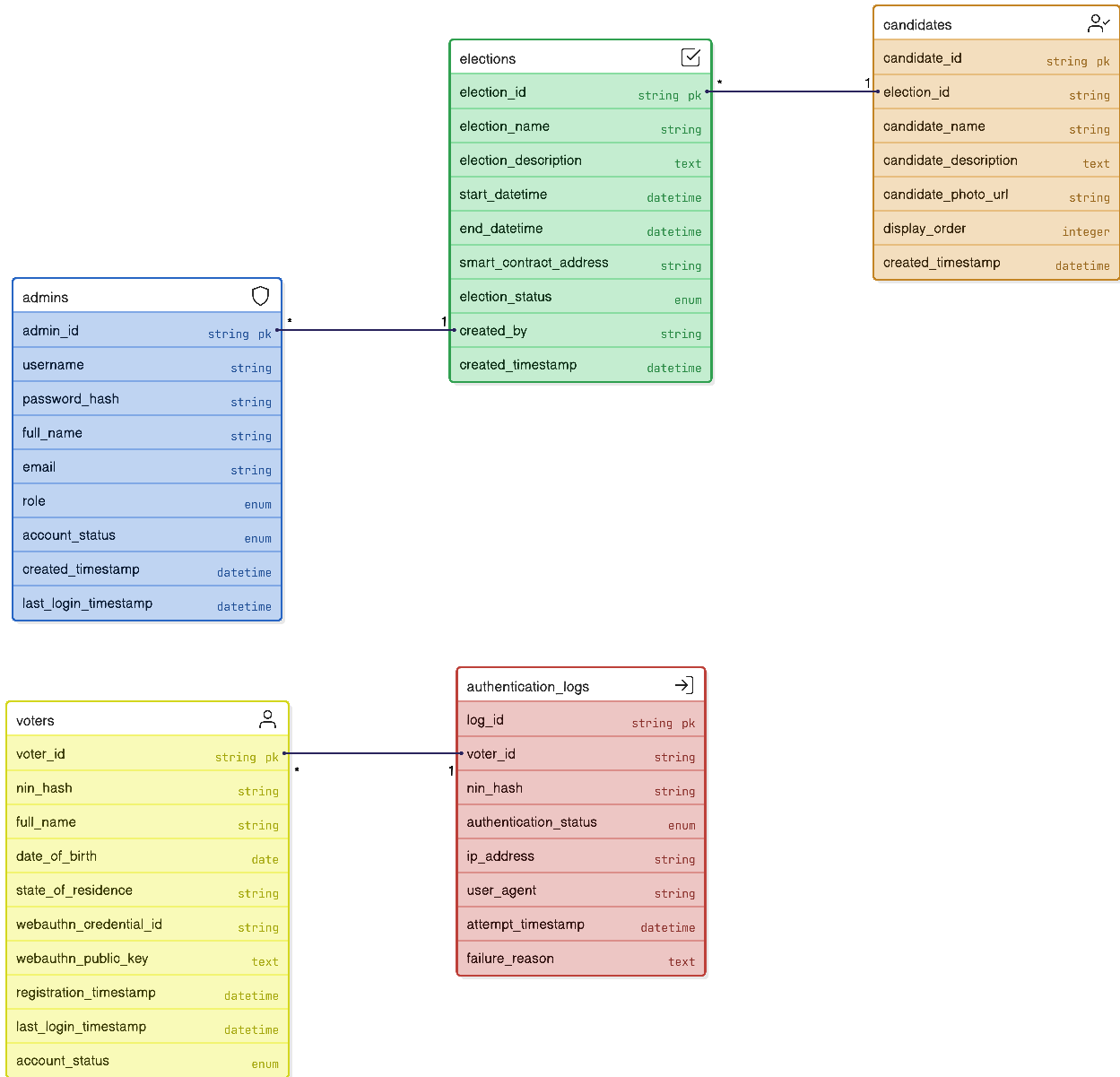


Figure 3.6: Entity-Relationship Diagram (ERD) showing the database tables / entities

2. VoterRegistry Smart Contract:

This contract maintains the registry of registered voters and their voting status.

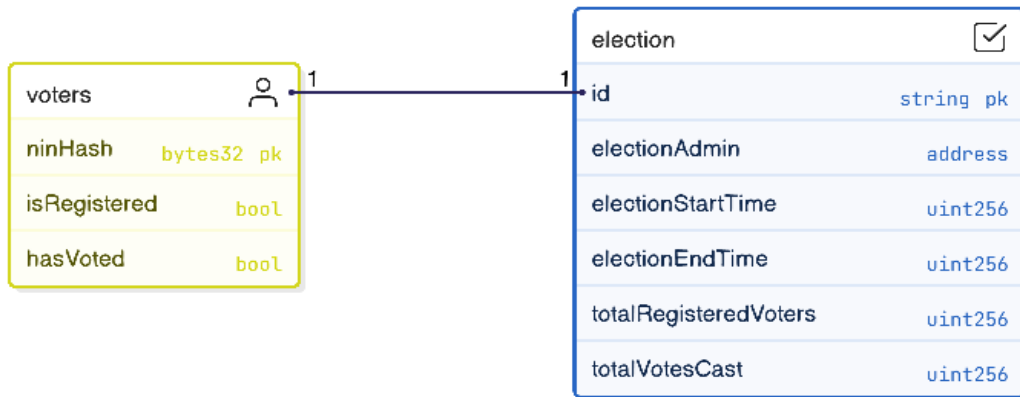


Figure 3.7: VoterRegistry Smart Contract Structure

3. VoteRecorder Smart Contract:

This contract handles immutable (Permanent) vote recording and tallying.

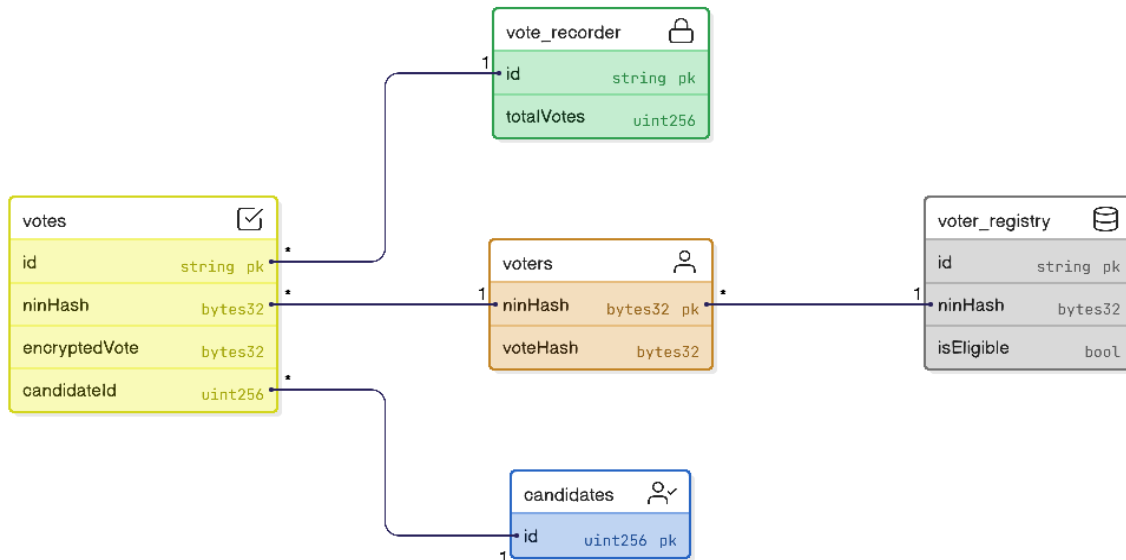


Figure 3.7: VoteRecorder Smart Contract Structure

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